



ARTI 506 - Data Science and Analytics

Term 2 – 2025/2026 - Assignment 1

Data-Driven Analysis of Tourism Demand and Spending Patterns in Saudi Arabia



AI Year 5

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1. Introduction

1.1 Introduction

Tourism in Saudi Arabia has become an important contributor to economic growth and development. As part of the Saudi Vision 2030, the tourism sector aims to diversify the economy and promote regional growth. The Kingdom has set a new goal to attract 150 million tourists by 2030 [1]. Understanding tourism patterns is crucial for strategic planning, as tourist demand varies by visitor type (inbound or domestic), destination, and purpose of the visit. A key issue is that the high number of visitors to a particular region does not necessarily mean the highest tourism spending, because spending patterns are influenced by the purpose of the visit. This project aims to address this issue by analyzing tourism datasets of domestic and inbound tourists to uncover patterns in visitor distribution and spending behavior across different regions and visit purposes in Saudi Arabia from 2015 to the first half of 2025. This analysis will provide insights to support the development of data-driven tourism strategies aligned with Saudi Vision 2030.

1.2 Problem Statement

Tourism is a significant factor that affects the economy of kingdom of Saudi Arabia. Tourist demand varies by visitor type (inbound or domestic) and by destination and purpose of visit. The main problem is high number of visiting a certain region doesn't mean the highest spending rate because of the purpose of the visits. This gap shows the difficulty in deciding where to focus new investments or how to market other cities differently to increase the tourism movement in Saudi Arabia. This project aims to analyze tourism datasets for both inbound and domestic to show patterns in visitor distribution and spending behavior from 2015-2025H1.

1.3 Objectives

This project aims to explore, merge, and analyze four datasets related to the tourism sector in Saudi Arabia. It seeks to provide a comparison between inbound and domestic tourists, generate insights into tourism trends, and forecast future tourism patterns. To predict future tourism, machine learning algorithms will be applied, while exploratory data analysis (EDA) will be conducted to identify trends and patterns within the data. Additionally, data visualization techniques will be utilized to facilitate user interaction with the EDA findings and the predictions generated by the machine learning models.



1.4 Dataset description & Justification

For our project, we are using four datasets from the Saudi Open Data Portal that track the development of the Kingdom tourism sector from 2015 to 2025 H1. These datasets include detailed records of inbound and domestic tourists by their main purpose (such as religious, leisure, or business) [2][3], as well as spending and arrival statistics broken down by specific provinces and regions [4][5]. We are using this data to study how tourism trends are changing. By looking at both visitors number and spending, we can see which trip purposes, like religious or leisure travel, contribute most to the economy and how those patterns differ between local and international tourists.

This project aligns with Saudi Vision 2030 which aims for 150 million visitors and a 10% GDP contribution from tourism by the end of the decade. Using ten years of data, we can apply predictive modeling to forecast demand and identify emerging destinations, providing actionable insights for sustainable planning.

1.5 Initial analysis of the dataset's potential

After going through the four tourism datasets from the Saudi Open Data Platform, we believe they have strong potential for our project. The data covers a long period from 2015 to the first half of 2025 and includes important details like visitor type (inbound and domestic), main purpose of visit, region (13 administrative regions), number of tourists, and their spending. This helps us look at tourism from more than one side, not just how many people visit, but also how much they actually spend, which is the main issue we are trying to study. Since the data is organized by year and region, we can easily merge the datasets and calculate useful indicators such as spending per tourist and growth over time. Because the data spans almost ten years, it also allows us to apply forecasting models to predict future tourism trends. Even though the data is yearly and doesn't show detailed seasonal patterns, it is official, reliable, and detailed enough to support strong exploratory analysis and machine learning models that align with Vision 2030 goals.



2. Data Acquisition

2.1 Data Loading

In this part, we load the raw data so we could work with it. We used pandas to read the four different CSV files, two about domestic (local) tourism and two about inbound (international) tourism. After loading them, we print the shape of each dataset, which is a quick way to see how many rows and columns we have for each dataset Figure 1, and we use head(3) command to preview the data, which displays the first three rows of each table Figure 2.

```
Domestic by Province : 143 rows, 6 columns
Domestic by Purpose : 55 rows, 6 columns
Inbound by Purpose : 55 rows, 6 columns
Inbound by Province : 141 rows, 6 columns
```

Figure 1 Dataset shape

```
--- Domestic by Province ---
TOURIST TYPE EN TRIP TYPE YEAR DESTINATION PROVINCE NAME EN Total VISITORS Number SPENDS by SR
0 DOMESTIC Overnight 2015 Albaha 1,041,399 1,119,776,018.55
1 DOMESTIC Overnight 2015 Alqassim 946,993 832,038,134.23
2 DOMESTIC Overnight 2015 Asseer 5,091,717 6,482,489,011.94

--- Domestic by Purpose ---
TOURIST TYPE EN TRIP TYPE YEAR VISIT PURPOSE GROUPED EN VISITORS SPENDS
0 DOMESTIC Overnight 2015 Business 1,859,752 3,247,666,761.42
1 DOMESTIC Overnight 2015 Leisure 19,181,792 21,183,455,240.20
2 DOMESTIC Overnight 2015 Other 2,335,328 4,059,351,265.38

--- Inbound by Purpose ---
TOURIST TYPE EN TRIP TYPE YEAR VISIT PURPOSE GROUPED EN VISITORS SPENDS
0 INBOUND Overnight 2015 Business 1,880,575 10,090,826,436.41
1 INBOUND Overnight 2015 Leisure 1,560,665 4,954,068,163.96
2 INBOUND Overnight 2015 Other 2,392,366 9,067,180,026.90

--- Inbound by Province ---
TOURIST TYPE EN TRIP TYPE YEAR DESTINATION PROVINCE NAME EN VISITORS SPENDS
0 INBOUND Overnight 2015 Albaha 6,092 42,822,952.65
1 INBOUND Overnight 2015 Alqassim 41,902 143,586,806.97
2 INBOUND Overnight 2015 Asseer 59,257 503,026,385.00
```

Figure 2 Preview first 3 rows of each dataset

2.2 Data Summary

Once the data was loaded, we wanted to see how it was structured. We use the .info() function to get a high-level summary Figure3. This showed us the names of every column and, more importantly, the data types helping us realize that many numbers were currently being treated as objects instead of actual numbers, which we will clean later.



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 143 entries, 0 to 142
Data columns (total 6 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   TOURIST TYPE EN                       143 non-null    object
1   TRIP TYPE                             143 non-null    object
2   YEAR                                  143 non-null    object
3   DESTINATION PROVINCE NAME EN         143 non-null    object
4   Total VISITORS Number                 143 non-null    object
5   SPENDS by SR                          143 non-null    object
dtypes: object(6)
memory usage: 6.8+ KB

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55 entries, 0 to 54
Data columns (total 6 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   TOURIST TYPE EN                       55 non-null    object
1   TRIP TYPE                             55 non-null    object
2   YEAR                                  55 non-null    object
3   VISIT PURPOSE GROUPED EN             55 non-null    object
4   VISITORS                             55 non-null    object
5   SPENDS                                55 non-null    object
dtypes: object(6)
memory usage: 2.7+ KB

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55 entries, 0 to 54
Data columns (total 6 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   TOURIST TYPE EN                       55 non-null    object
1   TRIP TYPE                             55 non-null    object
2   YEAR                                  55 non-null    object
3   VISIT PURPOSE GROUPED EN             55 non-null    object
4   VISITORS                             55 non-null    object
5   SPENDS                                55 non-null    object
dtypes: object(6)
memory usage: 2.7+ KB

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 141 entries, 0 to 140
Data columns (total 6 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   TOURIST TYPE EN                       141 non-null    object
1   TRIP TYPE                             141 non-null    object
2   YEAR                                  141 non-null    object
3   DESTINATION PROVINCE NAME EN         141 non-null    object
4   VISITORS                             141 non-null    object
5   SPENDS                                141 non-null    object
dtypes: object(6)
memory usage: 6.7+ KB
```

Figure 3 Datasets information

2.3 Statistical Description

To understand the data better, we use the `.describe()` command. This gave us a mathematical preview of the data Figure 4, showing things like the count of unique values and the most frequent entries. Since our numbers were stored as text, it helped us identify things like the range of years (2015 to 2025 H1) and the specific regions included in the study.

--- Domestic Province ---						
	TOURIST TYPE EN	TRIP TYPE	YEAR	DESTINATION PROVINCE NAME EN	Total VISITORS Number	SPENDS by SR
count	143	143	143	143	143	143
unique	1	1	11	13	143	143
top	DOMESTIC	Overnight	2015	Albaha	1,041,399	1,119,776,018.55
freq	143	143	13	11	1	1
--- Domestic Purpose ---						
	TOURIST TYPE EN	TRIP TYPE	YEAR	VISIT PURPOSE GROUPED EN	VISITORS	SPENDS
count	55	55	55	55	55	55
unique	1	1	11	5	55	55
top	DOMESTIC	Overnight	2015	Business	1,859,752	3,247,666,761.42
freq	55	55	5	11	1	1
--- Inbound Purpose ---						
	TOURIST TYPE EN	TRIP TYPE	YEAR	VISIT PURPOSE GROUPED EN	VISITORS	SPENDS
count	55	55	55	55	55	55
unique	1	1	11	5	55	55
top	INBOUND	Overnight	2015	Business	1,880,575	10,090,826,436.41
freq	55	55	5	11	1	1
--- Inbound Province ---						
	TOURIST TYPE EN	TRIP TYPE	YEAR	DESTINATION PROVINCE NAME EN	VISITORS	SPENDS
count	141	141	141	141	141	141
unique	1	1	11	13	141	141
top	INBOUND	Overnight	2015	Aqassim	6,092	42,822,952.65
freq	141	141	13	11	1	1

Figure 4 Description of each dataset



3. Data Cleaning

3.1 Cleaning Function

Instead of manually fixing every dataset one by one, we wrote a reusable cleaning function to automate the process. This function does three main things, it removes commas from numbers so the computer can do math with them, renames columns so they match across all four files, and converts the year column into a numeric format, for example, turning "2025 H1" into 2025.5.

```
--- Domestic Province ---
Shape      : 143 rows x 6 columns
Columns    : ['TOURIST TYPE EN', 'YEAR', 'DESTINATION PROVINCE NAME EN', 'VISITORS', 'SPENDS', 'YEAR_NUM']
Year range : 2015 - 2025 H1

--- Inbound Province ---
Shape      : 141 rows x 6 columns
Columns    : ['TOURIST TYPE EN', 'YEAR', 'DESTINATION PROVINCE NAME EN', 'VISITORS', 'SPENDS', 'YEAR_NUM']
Year range : 2015 - 2025 H1

--- Domestic Purpose ---
Shape      : 55 rows x 6 columns
Columns    : ['TOURIST TYPE EN', 'YEAR', 'VISIT PURPOSE GROUPED EN', 'VISITORS', 'SPENDS', 'YEAR_NUM']
Year range : 2015 - 2025 H1

--- Inbound Purpose ---
Shape      : 55 rows x 6 columns
Columns    : ['TOURIST TYPE EN', 'YEAR', 'VISIT PURPOSE GROUPED EN', 'VISITORS', 'SPENDS', 'YEAR_NUM']
Year range : 2015 - 2025 H1
```

Figure 5 Dataset cleaning function

3.2 Handling Missing Values

Before starting any deep analysis, we check for values in our data where information might be missing. We scanned every row and column in all four datasets for empty values. The result in Figure 6 showed 0 Missing Values for every file, which confirmed that our datasets were complete and ready for analysis.

```
--- Domestic (Province) ---
Missing Values: 0
-----
--- Inbound (Province) ---
Missing Values: 0
-----
--- Domestic (Purpose) ---
Missing Values: 0
-----
--- Inbound (Purpose) ---
Missing Values: 0
-----
```

Figure 6 Number of missing values

3.3 Outlier Identification

In this step, we look for extreme values that are much higher or lower than the rest. We use boxplots to visualize these outliers in Figure 7. We discovered that while there are extreme values, like very high visitor counts in Makkah and Riyadh, they aren't mistakes, they just represent the busiest regions. To stop these huge numbers from making our charts look messy, we capped them at the 95th percentile, which keeps the data honest without letting the extreme value hide the smaller details.

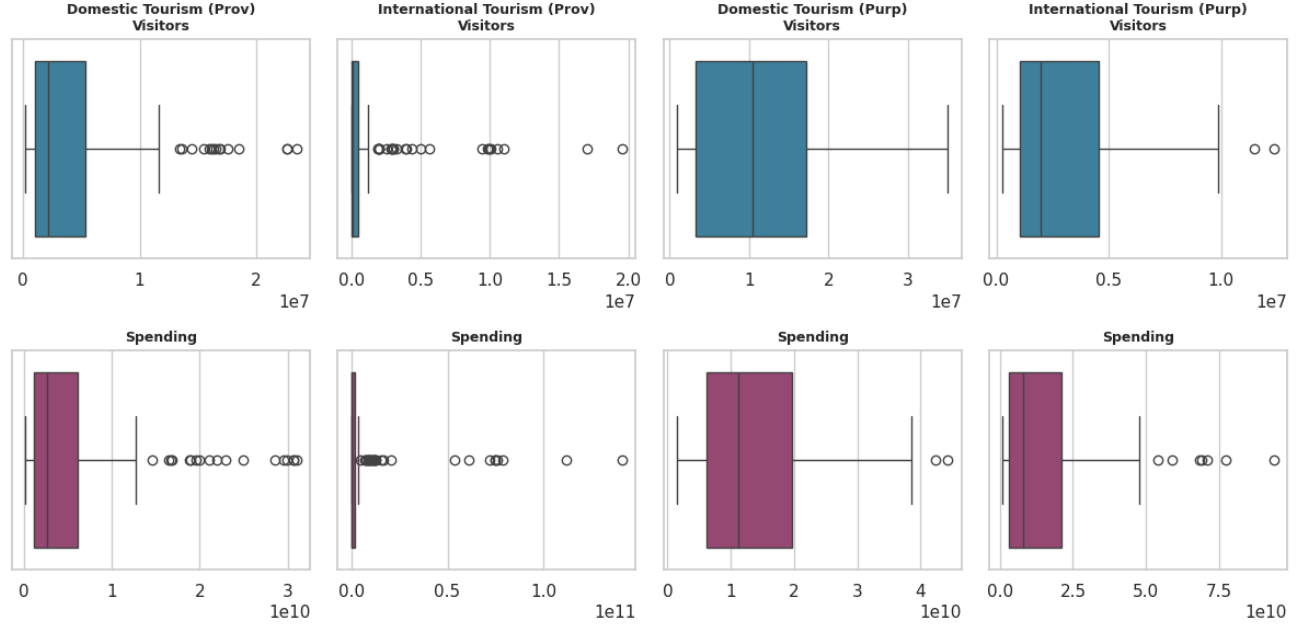


Figure 7 Outliers identification

4. Feature Engineering

4.1 Adding New Features

Utilizing existing data to generate new features is an essential step in data science and machine learning, as it enhances the ability of the model to capture useful patterns. In this project, a new feature, SPEND_PER_TOURIST, was created and added to all four datasets (Domestic Province, Inbound Province, Domestic Purpose, and Inbound Purpose). The feature can be derived by dividing SPENDS over VISITORS, it provides a measure of the average spending per tourist and allow to understand the individual spending behavior across different provinces and purposes, rather than relying on total spends. Figure 8 presents sample values from the Domestic Purpose dataset.

	YEAR	VISIT	PURPOSE	GROUPED EN	VISITORS	SPENDS	SPEND_PER_TOURIST
0	2015			Business	1859752.0	3.247667e+09	1746.290237
1	2015			Leisure	19181792.0	2.118346e+10	1104.352254
2	2015			Other	2335328.0	4.059351e+09	1738.236027
3	2015			Religious	11972476.0	1.015693e+10	848.356278
4	2015			VFR	11101085.0	9.771687e+09	880.246154

Figure 8 Sample values of 'SPEND_PER_TOURIST' feature.



4.2 Visualizing skewness and applying Log Transformation.

In the outlier identification stage, Figure 7 showed a clear right skewness in both visitor counts and total spending across all four datasets. This skewness resulted from several factors, such as Makkah having higher visitor numbers compared to other provinces, and inbound tourists spending more on specific purposes like religious and business travel. Skewed data can impact analysis and machine learning model performance negatively, and it may cause bias in the results due to extreme values. To address this issue, a log transformation ($\log(x + 1)$) was applied to both VISITORS and SPENDS across all datasets, and one was added to handle zero values. As shown in Figure 9, the original distribution of visitor data shows right skewness. After applying the log transformation, the distribution becomes balanced and close to a normal (bell-shaped) distribution. These transformed features will be utilized in later stages of the project as it suitable for machine learning models.

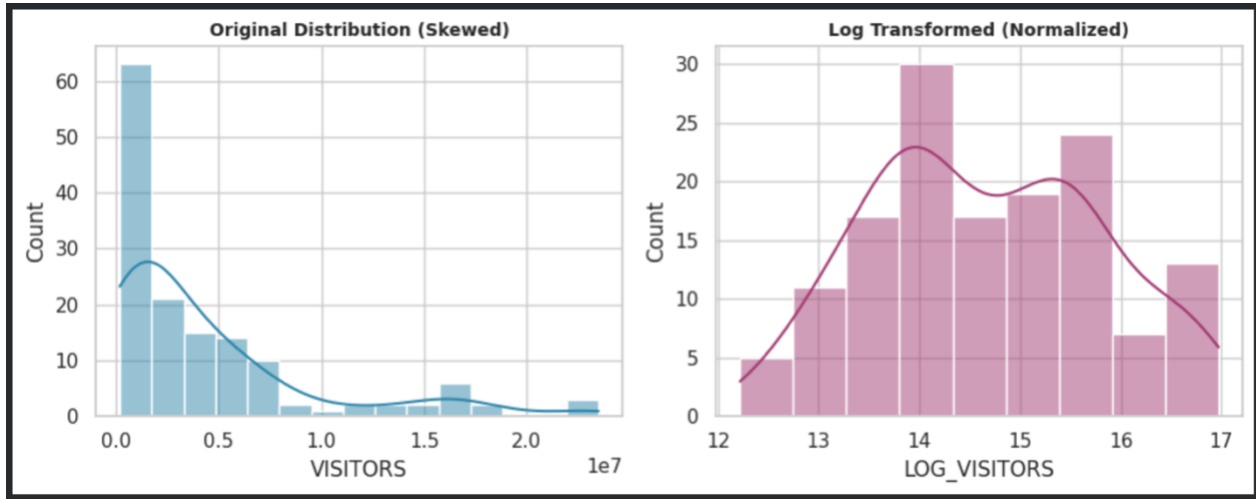


Figure 9: Visualization of skewed distribution vs. normal distribution.

5. Exploratory Data Analysis

5.1 Overall Tourism Growth Trends (2015–2025).

The Figure 10 illustrates the total number of tourists in Saudi Arabia from 2015 to 2025 H1, distinguishing between domestic tourism and international tourism. The x-axis represents the timeline, while the y-axis shows the total number of visitors. From the graph, we can see that domestic tourism consistently exceeds international tourism throughout the period. Both segments dropped sharply in 2020 due to the impact of COVID-19, followed by a strong recovery from 2021 onward, with tourism levels peaking around 2024 before a slight drop in 2025 H1. The 2025 H1 figure is lower simply because it only covers half the year. International tourism shows greater fluctuations compared to the more stable domestic trend.



Figure 10 Line chart showing total tourism growth in Saudi Arabia from 2015 to 2025 (H1)

5.2 Regional Visitor Trends Over Time.

The chart in the Figure 11 tracks how visitor numbers changed over time in the five most visited provinces in Saudi Arabia, from 2015 to 2025 H1. Makkah consistently attracts the highest number of tourists, followed by Riyadh and the Eastern Province with stable growth patterns, while Madinah and Aseer maintain lower but steady visitor numbers. The COVID-19 impact in 2020 is visible across all five regions, but recovery speed and peak levels differ, suggesting that regional tourism drivers are not uniform and that targeted regional strategies would be more effective than a one-size-fits-all approach.

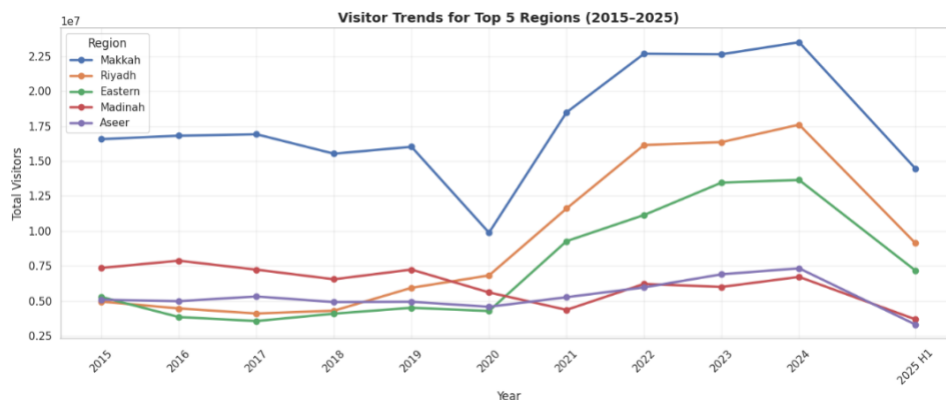


Figure 11 multi-line chart illustrating visitor trends over time for the top five regions in Saudi Arabia

5.3 Visitor Volume vs. Spending Shares (Market Share Analysis).

These two pie charts in Figure 12 compare the percentage of total visitors and total spending that comes from domestic versus international tourists. Domestic tourists make up 77.6% of all visitors but contribute only 44.7% of total spending. International tourists are only 22.4% of visitors but generate 55.3% of spending. This highlights the importance of attracting international tourists as a key driver of tourism revenue, while domestic tourism remains essential for maintaining overall visitor numbers and market stability.

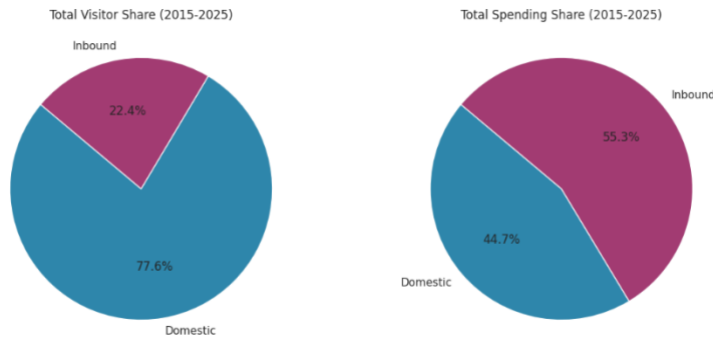


Figure 12 Pie charts showing visitor and spending shares for domestic and inbound tourism in Saudi Arabia

5.4 Comparing Average Spending per Tourist

The Figure 13 shows the trend in average tourist spending using SAR for both domestic and international tourists from 2015 to 2025. International tourists consistently spend more than domestic tourists, with values fluctuating but generally increasing over time. In contrast, domestic tourist spending remains relatively stable with only slight variations. Both dipped during COVID, but international spend per tourist has reached its highest level ever at around 6,300 SAR in 2025 H1, while domestic spend per tourist has remained largely flat between 1,000 and 1,500 SAR throughout the entire period. Overall, International tourists contribute higher spending per visitor, while domestic spending shows steady but lower levels.

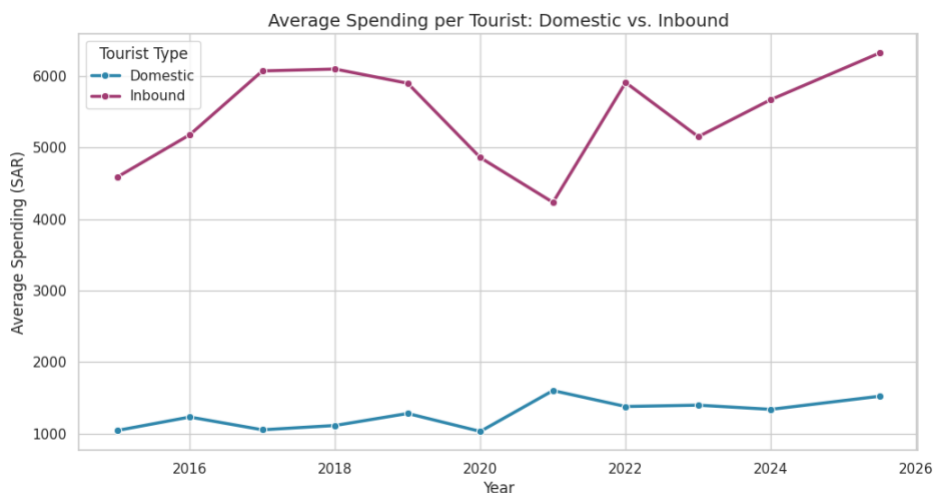


Figure 13 Line chart showing the average spending per tourist (SAR) for domestic and inbound tourists from 2015 to 2025.



5.5 Comparing Inbound vs. Domestic Travel Purposes.

This Bar chart in Figure 14 compares the number of domestic and international visitors by purpose of visiting. Leisure is the most common purpose for domestic tourists, followed by religious visits, then VFR (visiting friends and relatives), which is a personal trip type associated with lower spending, helping explain why domestic tourists spend less per person. On the other hand, international tourists are highest in religious visits, with more evenly spread across the remaining purposes compared to domestic tourists.

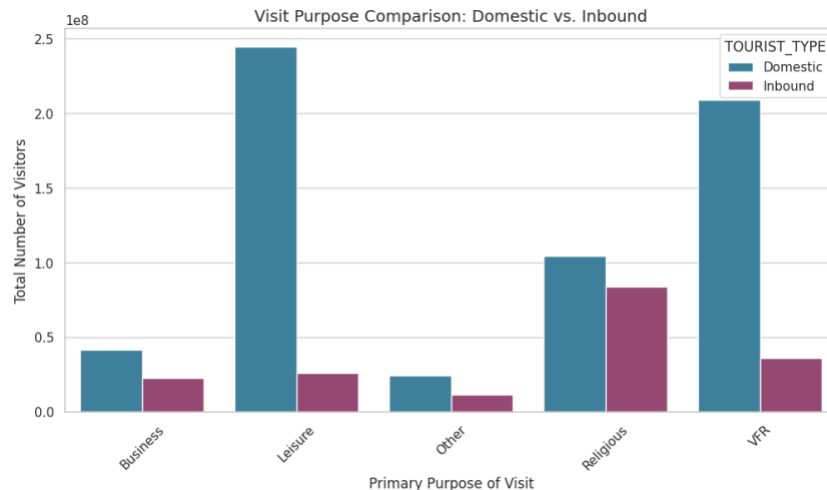


Figure 14 Bar chart comparing domestic and inbound visitors by primary purpose of visit

5.6 Tourism by Destination Comparison (Regional Performance).

This chart in Figure 15 shows which provinces attract the most tourists, comparing domestic and international visitors side by side across all 13 regions. Makkah has the highest number of visitors in both categories. For domestic tourism, Riyadh and the Eastern Province also have high visitor numbers, followed by Madinah and Aseer, while other provinces attract significantly fewer tourists. For international tourists, Makkah alone accounts for most arrivals, confirming that international tourism is heavily driven by religious visits to the holy cities.

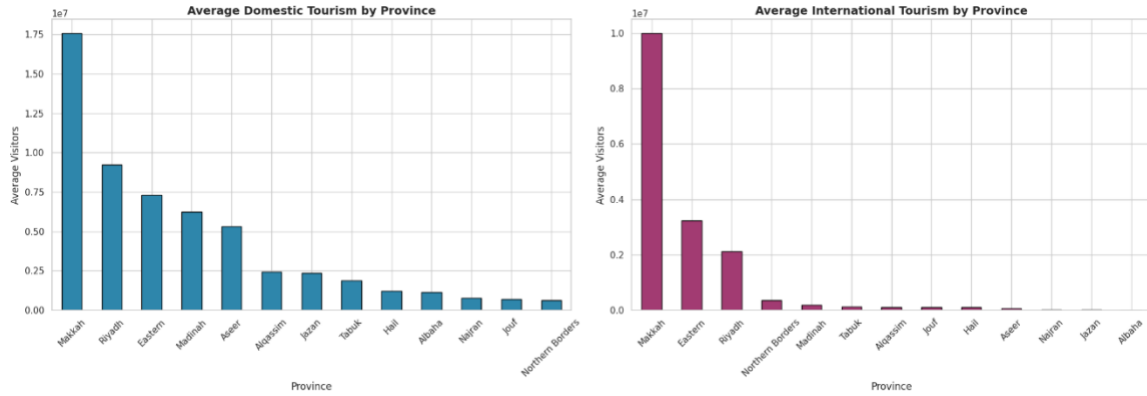


Figure 15 Bar charts illustrating the average number of domestic and international tourists by province.

6. Advanced Insights & Correlation

6.1 Feature Selection

Feature selection is an essential step in data analysis and machine learning, where the most relevant features are selected to improve model performance and reduce the impact of irrelevant variables. In this project, seven key features were selected based on their importance in representing tourism patterns. One of the primary features is TOURIST_TYPE, which distinguishes between domestic and inbound tourists. Since the datasets are structured based on both provinces and purposes, DESTINATION_PROVINCE_NAME_EN and VISIT_PURPOSE_GROUPED_EN were also included. The feature YEAR_NUM represents the temporal dimension of the data, containing 11 unique values spanning from 2015 to the first half of 2025. This wide time range enables the analysis of trends over time and supports forecasting future tourism patterns. Additionally, log-transformed features, LOG_VISITORS and LOG_SPENDS, were computed to reduce data skewness and improve model stability. These transformations help machine learning models better capture patterns. Finally, SPEND_PER_TOURIST was included as a derived feature to represent individual spending behavior. This feature provides insights into how much each tourist spends across different regions and purposes, enhancing the ability of the model to analyze patterns in tourism.

6.2 Apply Correlation

In this section, Pearson correlation is utilized to measure and analyze the relationships between different numerical features across the four datasets. The correlation coefficient ranges from -1 to +1, where a value close to +1 indicates a strong positive relationship between two features, while a value close to -1 indicates a strong negative relationship. A value near 0 suggests that there is no linear relationship between the variables [6].



6.2.1 Relationship Between Visitor Volume and Total Spending

This section aims to examine the relationship between visitor volume and total tourism spending. A scatter plot was used to visualize the relationship between the number of visitors and total spending for both domestic and inbound tourism datasets. In addition, a linear trend line was applied to identify the overall direction of the relationship, along with Pearson correlation to quantify its strength. As shown in Figure 16, the results indicate a strong positive relationship between visitor numbers and total spending for both domestic and inbound tourists, with a correlation coefficient of 0.97. This suggests that increases in visitor volume are strongly associated with higher tourism revenue. However, the data points are scattered around the trend line, indicating that visitor numbers alone do not fully determine total spending. Some provinces may attract a high number of visitors while generating lower spending per tourist. This supports the idea that visitor quantity and economic contribution are not always directly proportional.

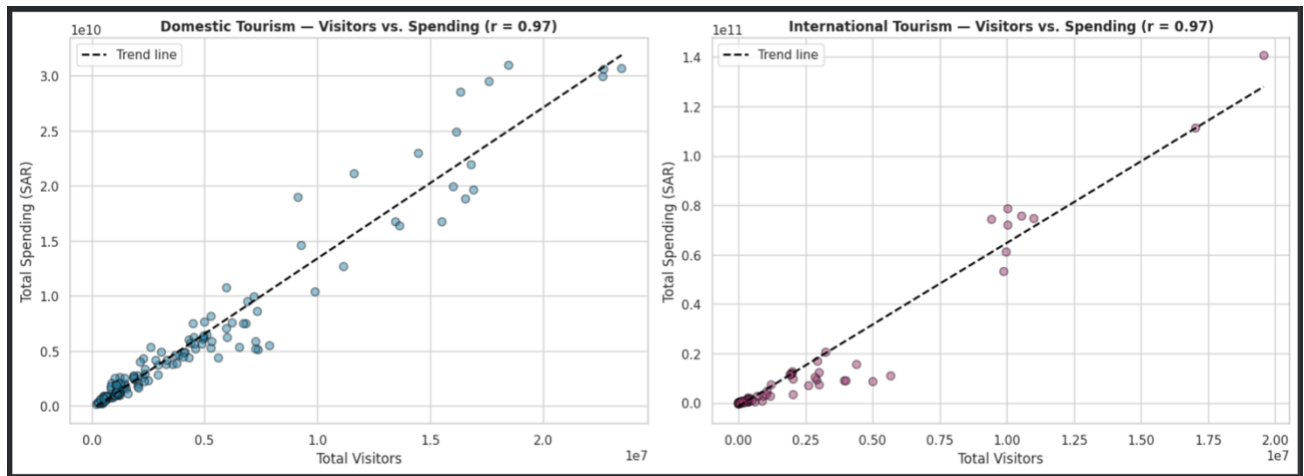


Figure 16: Scatterplot of visitors vs. spending for both domestic and inbound tourism.

6.2.2 Feature Correlation Heatmaps Across All Datasets

This section examines the relationships between the main numerical features across the four tourism datasets using correlation heatmaps. The analysis focuses on YEAR_NUM, VISITORS, SPENDS, and SPEND_PER_TOURIST features. Correlation matrices were computed and visualized using heatmaps as shown in Figure 17, where red indicates a strong positive relationship, blue represents a negative relationship, and lighter colors indicate weak or no relationship. Across all datasets, a strong positive correlation was observed between VISITORS and SPENDS, with values ranging between 0.96 and 0.97. This indicates that higher visitor volumes are strongly associated with increased total tourism spending. The relationship between VISITORS and SPEND_PER_TOURIST varies across datasets. In most cases, the correlation is weak to moderate. However, an exception appears in the Domestic Tourism by Purpose dataset, where a negative correlation of -0.42 is



observed. This suggests that the purposes attracting the highest number of domestic tourists tend to generate lower spending per individual. Similarly, the relationship between SPENDS and SPEND_PER_TOURIST differs by tourism type. In inbound tourism by purpose, a relatively strong positive correlation of 0.64 is observed, indicating that higher total spending is associated with increased spending per tourist. In contrast, domestic tourism by purpose shows a weak negative relationship of -0.20, suggesting a different spending pattern. Finally, the relationship between YEAR_NUM and SPEND_PER_TOURIST shows a moderate positive trend in domestic tourism, indicating a gradual increase in spending per tourist over time. In contrast, inbound tourism exhibits only a weak relationship, suggesting that spending behavior among international tourists remains relatively stable over the years.

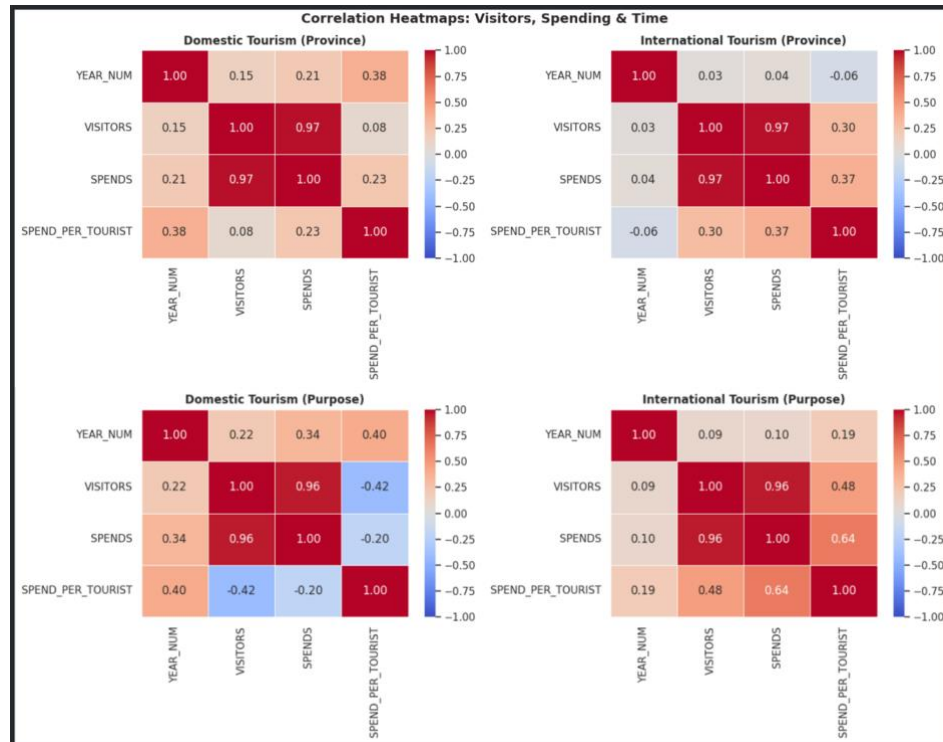


Figure 17: Correlation heatmaps across all datasets for visitors, spending, and time.



6.2.3 Pearson Correlation Summary

This section provides a summary of the correlation results identified across all four datasets. A summary table was created to recognize the key relationships between the four selected numerical features, offering a clearer comparison of correlation patterns observed in the heatmaps. The table categorizes correlation using color coding by identifying strong correlations (≥ 0.7) in green, moderate correlations (≥ 0.4) in yellow, and weak correlations are left plain. As shown in Figure 18, VISITORS and SPENDS show the strongest correlation across all datasets, confirming that tourism revenue is dependent on the number of visitors. In contrast, features associated with time (YEAR_NUM) show weak correlations, indicating that tourism growth does not follow a linear trend over the years and is influenced by external events such as COVID in 2020. An exception is observed in the relationship between SPEND_PER_TOURIST and YEAR_NUM for domestic tourism by purpose, where a moderate correlation suggests a gradual increase in individual spending over time.

	Visitors vs Spending	Visitors vs Year	Spending vs Year	Spend/Tourist vs Year
Dataset				
Domestic Tourism (Province)	0.970000	0.150000	0.210000	0.380000
International Tourism (Province)	0.970000	0.030000	0.040000	-0.060000
Domestic Tourism (Purpose)	0.960000	0.220000	0.340000	0.400000
International Tourism (Purpose)	0.960000	0.090000	0.100000	0.190000

Figure 18: Pearson correlation summary table.

7. Conclusion

Tourism in Saudi Arabia has become an important factor influencing the economic sector. However, an increase in the number of visitors to a specific province or for a specific purpose does not necessarily reflect an increase in tourism revenue. In this report, four datasets (Domestic Province, Inbound Province, Domestic Purpose, and Inbound Purpose) were loaded and explored. Data cleaning was applied, and outliers were handled by capping them at the 95th percentile. In addition, a new feature '*SPEND_PER_TOURIST*' was added to measure the average spending per tourist, and skewed data was addressed using log transformation. Furthermore, exploratory data analysis was conducted to examine tourism trends, including total tourism over the years, the most attractive provinces, comparisons between domestic and inbound tourists, and spending behavior across different purposes. For further analysis, seven features were selected, and Pearson correlation was applied to the numerical ones. The results showed a strong positive relationship between visitors and spending across all datasets. However, domestic tourism by purpose showed lower spending per tourist, and overall, spending per tourist did not show consistent improvement over the years. In the next phase, a machine learning model will be developed to forecast future tourism and provide further insights.



8. References

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Appendix

Code link in GitHub

<https://github.com/faihadiln-spec/DS.git>

Power BI Dashboard Link

https://udksa-my.sharepoint.com/:u:/g/personal/2220002599_iau_edu_sa/IQBOuJxROsp0T4v-Y_V-7pktAWt24XCm4RL-vTG1OdPzgX8?e=vrOIsX